HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **LLM-powered agents** for financial analysis using **LangGraph**, enabling automated market research and strategy hypothesis generation.
- Developed a **cognitive memory system** that connects market context, trading decisions, and historical outcomes, storing and retrieving data via a **PostgreSQL** backend and integrating with time-series analysis modules.
- Built a **multi-agent system** orchestrating various AI agents for quantitative strategy development, incorporating feedback loops for continuous refinement and performance analytics.
- Engineered **Python** backend services with **FastAPI** to serve AI models and APIs, handling approximately **1,500 requests per minute** during peak loads and managing secure REST endpoints.
- Integrated **vector databases** (e.g., Pinecone, Chroma) for efficient similarity search on financial documents and market data, supporting RAG pipelines for LLM applications.
- Containerized AI applications using **Docker** and deployed them to a cloud environment, establishing basic **CI/CD pipelines** with **Git** for streamlined development and release cycles.
- Implemented **authentication** and **role-based access control** for a multi-tenant SaaS platform, ensuring secure API access and data segregation for different user groups.

SKILLS

AI/ML – LLMs, Multi-Agent Systems, RAG, Embeddings, Agent Orchestration, LangGraph, LlamaIndex, FastAPI, scikit-learn, pandas, NumPy

Backend Development – Python, TypeScript, Node.js, FastAPI, REST APIs, WebSockets, Microservices

Databases – PostgreSQL, Vector Databases (e.g. Pinecone, Chroma), SQL

DevOps & Cloud – Docker, Git, CI/CD, Cloud Deployment (AWS concepts)

Financial Analysis & Trading – Quantitative Strategy, Backtesting, Time-Series Analysis, Risk Metrics, Broker APIs (concepts)

PROJECTS

AI Strategy Intelligence Platform

- Developed a **Python**-based AI strategy intelligence platform integrating **LLM** capabilities for market analysis and idea generation, supporting quantitative strategy development.
- Implemented a **RAG** pipeline that ingests and queries financial news and reports using **embeddings** and a **vector database**, providing context-aware insights to downstream AI agents.
- Designed REST APIs using **FastAPI** to expose strategy performance metrics and facilitate interaction with a **cognitive memory** component, enabling iterative strategy validation.

Multi-Agent System for Trading Analytics

- Built a multi-agent system using **LangGraph** to automate the process of backtesting trading strategies, incorporating market data feeds and performance evaluation logic.
- Engineered data processing pipelines in **Python** with **pandas** and **NumPy** to prepare historical financial time-series data for analysis and model training.
- Developed a PostgreSQL schema to store strategy parameters, trade executions, and risk metrics, supporting detailed analytics and out-of-sample testing.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)